UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 14

Fall 2025

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Theorem 1. *Given convex sets $\mathcal{C}_1, \ldots, \mathcal{C}_m$, their intersection $\mathcal{C}_1 \cap \mathcal{C}_2 \cap \cdots \cap \mathcal{C}_m$ is convex.*

Given vectors $\boldsymbol{a}^{(1)}, \ldots, \boldsymbol{a}^{(m)} \in \mathbb{R}^n$ and scalars $b_1, \ldots, b_m$, consider the half-spaces:

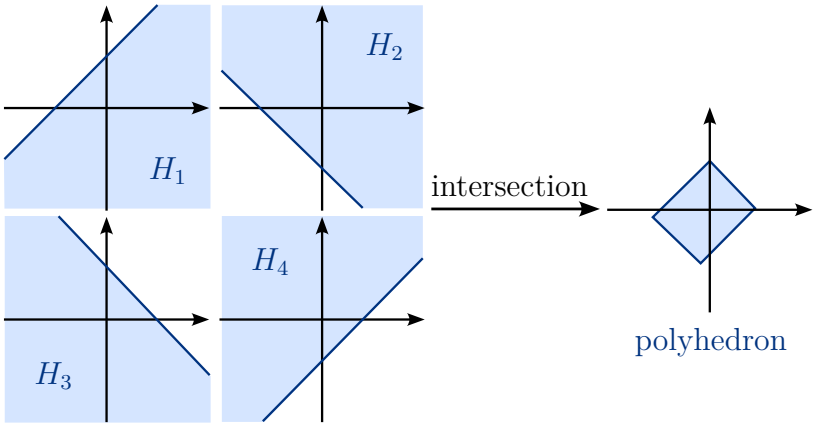
$$H_i = \left\{ \boldsymbol{x} \in \mathbb{R}^n \;\middle|\; \left(\boldsymbol{a}^{(i)}\right)^{\mathrm{T}} \boldsymbol{x} \leq b_i \right\}, \; i = 1, \ldots, m.$$

Since half-spaces are convex, by Theorem 1, their intersection is also convex:

$$H_1 \cap H_2 \cap \cdots \cap H_m = \left\{ \boldsymbol{x} \in \mathbb{R}^n \;\middle|\; \left(\boldsymbol{a}^{(1)}\right)^{\mathrm{T}} \boldsymbol{x} \leq b_1, \ldots, \left(\boldsymbol{a}^{(m)}\right)^{\mathrm{T}} \boldsymbol{x} \leq b_m \right\}.$$

This set is called a *polyhedron*. It has applications in linear program which are to be discussed later.

Example 1 (Half-spaces and polyhedron)**.**



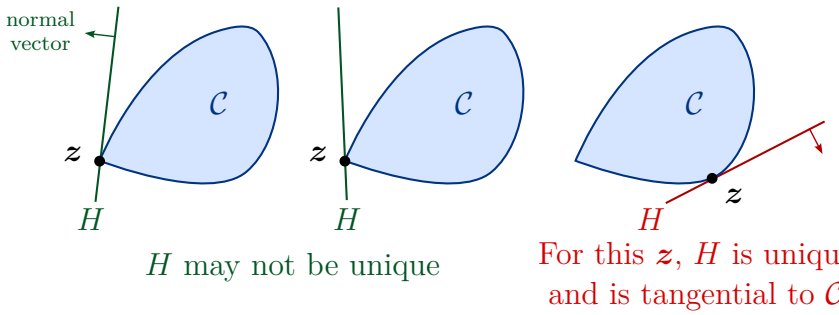
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Definition 1 (Supporting hyperplane)**.** Given a convex set $\mathcal{C}$ and a boundary point $\boldsymbol{z} \in \partial\mathcal{C}$, the hyperplane $H = \{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} = b\}$ is called a *supporting hyperplane* for $\mathcal{C}$ at $\boldsymbol{z}$ if:

(i) $\boldsymbol{z} \in H$ (i.e., $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{z} = b$);

(ii) $\mathcal{C} \subseteq H_-$ (where $H_- = \{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} \leq b\}$), meaning that $\boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} \leq b$ for all $\boldsymbol{x} \in \mathcal{C}$.

Example 2 (Supporting hyperplanes)**.** For a given boundary point $\boldsymbol{z}$, the supporting hyperplane at $\boldsymbol{z}$ may or may not be unique:



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The following theorem states that the supporting hyperplane *always exists*:

Theorem 2 (Supporting hyperplane theorem)**.** *If $\mathcal{C} \subseteq \mathbb{R}^n$ is a convex set and $\boldsymbol{z}$ is a point on the boundary of $\mathcal{S}$, then there exists a supporting hyperplane containing $\boldsymbol{z}$.*

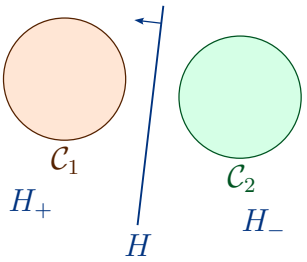
Definition 2 (Separating hyperplane)**.** Given two convex sets $\mathcal{C}_1, \mathcal{C}_2 \subseteq \mathbb{R}^n$, a hyperplane $H = \{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} = b\}$ is said to *separate* $\mathcal{C}_1$ and $\mathcal{C}_2$ if:

(i) $\mathcal{C}_1 \subseteq H_+$ (where $H_+ = \{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} \geq b\}$);

(ii) $\mathcal{C}_2 \subseteq H_-$ (where $H_- = \{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} \leq b\}$).

If $H \cap \mathcal{C}_1 = \varnothing$ and $H \cap \mathcal{C}_2 = \varnothing$, we say H *strictly separates* $\mathcal{C}_1$ and $\mathcal{C}_2$.

Example:



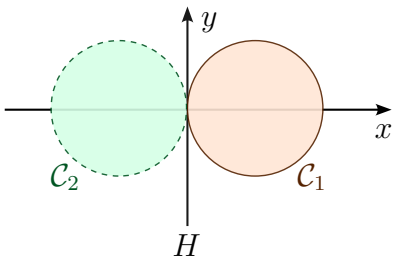
Theorem 3 (Separating hyperplane theorem)**.** *Let $\mathcal{C}_1, \mathcal{C}_2 \subseteq \mathbb{R}^n$ be convex sets.*

(i) If $\mathcal{C}_1 \cap \mathcal{C}_2 = \varnothing$, then a separating hyperplane of $\mathcal{C}_1$ and $\mathcal{C}_2$ exists.

(ii) If $\mathcal{C}_1 \cap \mathcal{C}_2 = \varnothing$, $\mathcal{C}_1$ and $\mathcal{C}_2$ are closed, and at least one of them is bounded, then a strictly separating hyperplane of $\mathcal{C}_1$ and $\mathcal{C}_2$ exists.

Example 3 (Separating hyperplanes)**.**

(a) Let $\mathcal{C}_1$ be a closed ball and $\mathcal{C}_2$ be an open ball:

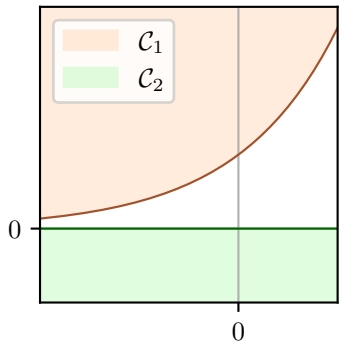


The vertical axis is the only separating hyperplane. Since $H \cap \mathcal{C}_1 = \{\mathbf{0}\} \neq \varnothing$, there is no strict separation.

(b) The following example shows that the requirement in (ii) of Theorem 3 that at least one of $\mathcal{C}_1$ and $\mathcal{C}_2$ being bounded is necessary:

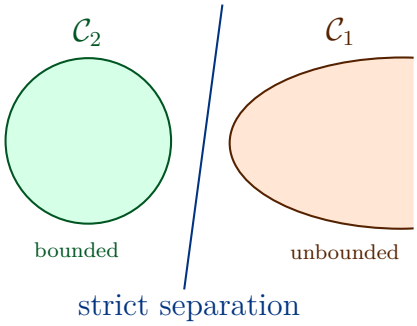
$$\mathcal{C}_1 = \{\boldsymbol{x} \in \mathbb{R}^2 \mid x_2 \geq \mathrm{e}^{x_1}\}, \quad \mathcal{C}_2 = \{\boldsymbol{x} \in \mathbb{R}^2 \mid x_2 \leq 0\}.$$

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We have $\mathcal{C}_1 \cap \mathcal{C}_2 = \varnothing$, and both $\mathcal{C}_1$ and $\mathcal{C}_2$ are closed. The only separating hyperplane H is the horizontal axis. But since $H \subseteq \mathcal{C}_2$, there is no strict separation.

(c) Therefore, we need boundedness of either $\mathcal{C}_1$ or $\mathcal{C}_2$ in addition to closedness to guarantee strict separation:



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Definition 3 (Domain)**.** The *domain* of a function $f\colon \mathbb{R}^n \to \mathbb{R}$ is the set of points $\boldsymbol{x}$ at which $f(\boldsymbol{x})$ is finite:

$$\operatorname{dom} f = \{\boldsymbol{x} \in \mathbb{R}^n \mid -\infty < f(\boldsymbol{x}) < +\infty\}.$$

Example 4 (Domain)**.**

(a) Let $f(x) = x^2$, then $\operatorname{dom} f = \mathbb{R} = (-\infty, +\infty)$.

(b) Let $f(x) = \log x$, then $\operatorname{dom} f = (0, +\infty)$.

(The logarithm of a negative number is a complex number and not within $-\infty$ and $+\infty$, i.e., undefined.)

(c) Let

$$f(\boldsymbol{x}) = \frac{\boldsymbol{a}^{\mathrm{T}}\boldsymbol{x} + b}{\boldsymbol{c}^{\mathrm{T}}\boldsymbol{x} + d}$$

where $\boldsymbol{a}, \boldsymbol{c} \in \mathbb{R}^n$ and $b, d \in \mathbb{R}$, then $\operatorname{dom} f = \{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{c}^{\mathrm{T}}\boldsymbol{x} + d \neq 0\}$.

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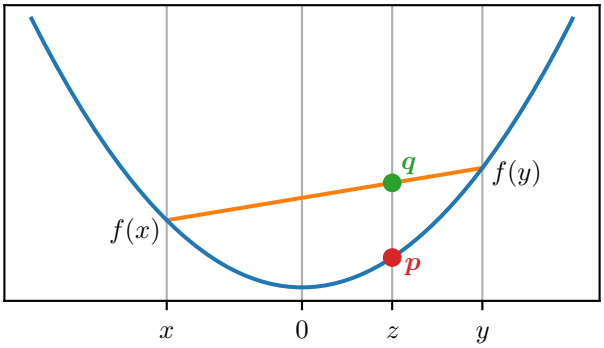
Definition 4 (Convex function)**.** A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is said to be *convex* if:

(i) $\operatorname{dom} f$ is a convex set;

(ii) For any $\boldsymbol{x}, \boldsymbol{y} \in \operatorname{dom} f$ and any $\alpha \in [0,1]$,

$$f\big(\alpha \boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) \le \alpha f(\boldsymbol{x}) + (1-\alpha) f(\boldsymbol{y}). \tag{1}$$

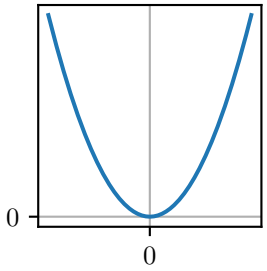
Interpretation of (1):



- Pick a point between x and y: $z = \alpha x + (1-\alpha)y$.
- The height of point $\boldsymbol{p}$ is $f(z)$.
- The height of point $\boldsymbol{q}$ is $\alpha f(x) + (1-\alpha) f(y)$.
- Convexity of f means that $\boldsymbol{q}$ is above $\boldsymbol{p}$, or equivalently, the segment connecting $(x, f(x))$ and $(y, f(y))$ is above the function f.

Example 5 (Convex functions)**.** Example of convex functions for $n = 1$:

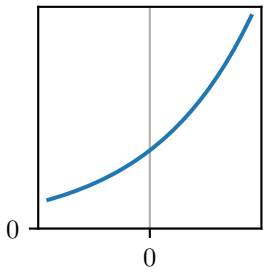
(a) $f(x) = x^2$:



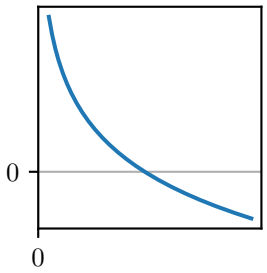
We can check that $\operatorname{dom} f = \mathbb{R}$ is a convex set, and f satisfies (1).

(b) $f(x) = \mathrm{e}^x$:

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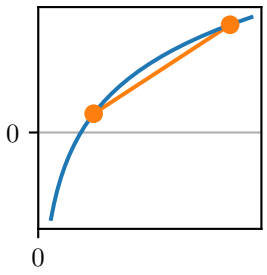


(c) $f(x) = -\log x$:



We can check that $\operatorname{dom} f = (0, +\infty)$ is a convex set, and that f satisfies (1).

Note: $\log x$ is not convex:



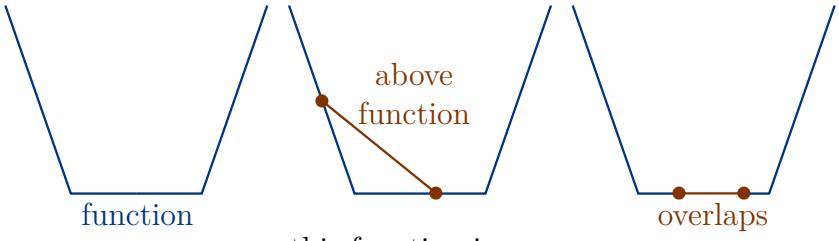
The segment shown is below the function, not above.

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Definition 5 (Concave function)**.** A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is *concave* if $-f(\boldsymbol{x})$ is convex.

As shown in Example 5(c), $\log x$ is a concave function.

Note: The inequality in (1) is not strict, so the segment can overlap with the function:



$\Rightarrow$ this function is convex

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If the segment is always strictly above the function, then $f(\boldsymbol{x})$ is called *strictly convex*:

Definition 6 (Strictly convex function)**.** A function $f\colon \mathbb{R}^n \to \mathbb{R}$ is *strictly convex* if:

(i) $\operatorname{dom} f$ is a convex set;

(ii) For any $\boldsymbol{x}, \boldsymbol{y} \in \operatorname{dom} f$ such that $\boldsymbol{x} \neq \boldsymbol{y}$ and any $\alpha \in (0,1)$,

$$f\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) < \alpha f(\boldsymbol{x}) + (1-\alpha) f(\boldsymbol{y}).$$

Theorem 4. *Let $f(\boldsymbol{x})$ be an arbitrary norm $\|\cdot\|\colon \mathbb{R}^n \to \mathbb{R}$ (defined in Definition 5 of Lecture Note 2). Then $f(\boldsymbol{x})$ is convex. (This is true for matrix norms too.)*

Proof. We have $\operatorname{dom} f = \mathbb{R}^n$ since the norm is defined for all $\boldsymbol{x} \in \mathbb{R}^n$. Also,

$$\underbrace{\|\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\|}_{f(\alpha\boldsymbol{x}+(1-\alpha)\boldsymbol{y})} \leq \|\alpha\boldsymbol{x}\| + \|(1-\alpha)\boldsymbol{y}\| = \alpha \underbrace{\|\boldsymbol{x}\|}_{f(\boldsymbol{x})} + (1-\alpha)\underbrace{\|\boldsymbol{y}\|}_{f(\boldsymbol{y})},$$

where we used the triangle inequality and $|\alpha| = \alpha,\ |1-\alpha| = 1-\alpha$ since $0 \leq \alpha \leq 1$. ■

Examples: $\|\boldsymbol{x}\|_2$, $\|\boldsymbol{x}\|_1$, $\|\boldsymbol{x}\|_\infty$ are all convex.

Theorem 5. *Given convex functions $f_1(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x})$ from $\mathbb{R}^n$ to $\mathbb{R}$, and given constants $w_1, \ldots, w_m$ such that $w_1, \ldots, w_m \geq 0$. Consider the function*

$$f(\boldsymbol{x}) = w_1 f_1(\boldsymbol{x}) + \cdots + w_m f_m(\boldsymbol{x}).$$

Then, the function $f(\boldsymbol{x})$ is convex with the domain

$$\operatorname{dom} f = \operatorname{dom} f_1 \cap \cdots \cap \operatorname{dom} f_m.$$

(Note: $f(\boldsymbol{x})$ is well-defined only when all functions $f_1(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x})$ have finite values, and that's why we take intersection of domains of $f_1(\boldsymbol{x}), \ldots, f_m(\boldsymbol{x})$.)

Proof. Since $f_1, \ldots, f_m$ are convex, $\operatorname{dom} f_1, \ldots, \operatorname{dom} f_m$ are convex sets. By Theorem 1, their intersection is also convex. Thus, $\operatorname{dom} f$ is a convex set.
Furthermore, by the convexity of $f_i(\boldsymbol{x})$, for any $\boldsymbol{x}, \boldsymbol{y} \in \operatorname{dom} f$, we have

$$f_i\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) \leq \alpha f_i(\boldsymbol{x}) + (1-\alpha) f_i(\boldsymbol{y}), \quad i = 1, \ldots, m.$$

Since $w_i \geq 0$, we can multiply both sides by w_i:

$$w_i f_i\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) \leq \alpha w_i f_i(\boldsymbol{x}) + (1-\alpha) w_i f_i(\boldsymbol{y}), \quad i = 1, \ldots, m.$$

Taking the sum for $i = 1, \ldots, m$, we get

$$\begin{aligned} &\sum_{i=1}^m w_i f_i\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) \leq \alpha \sum_{i=1}^m w_i f_i(\boldsymbol{x}) + (1-\alpha) \sum_{i=1}^m w_i f_i(\boldsymbol{y}) \\ \Rightarrow\ &f\big(\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{y}\big) \leq \alpha f(\boldsymbol{x}) + (1-\alpha) f(\boldsymbol{y}). \end{aligned}$$

Therefore, f is convex. ■

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Example 6. Since $x^2, x^4, -\log x, \mathrm{e}^x, \mathrm{e}^{-x}$ are convex functions, it follows that

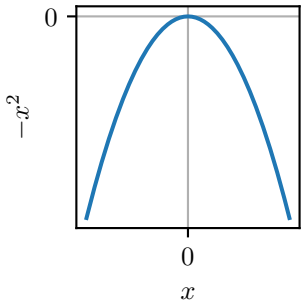
$$2.5x^2 + 4x^2 - 10\log x + 20\mathrm{e}^x + 20\mathrm{e}^{-x}$$

is also convex with domain $(0, +\infty)$. ■

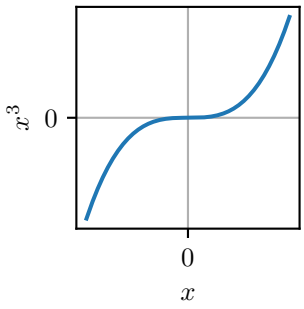
Set $m = 2$ and $w_1 = w_2 = 1$ in Theorem 5, we see that if f_1 and f_2 are convex, then $f_1 + f_2$ is also convex. Hence, addition preserves convexity.
However, subtraction, multiplication and division may not preserve convexity:

Example 7.

(a) 0 and x^2 are convex, but $0 - x^2 = -x^2$ is non-convex:



(b) x and x^2 are convex, but $x \cdot x^2 = x^3$ is non-convex:



(c) x^4 and x are convex, but $x^4/x = x^3$ is non-convex.

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